

Second Terminal Examination - 2079

Class : XI

Subject: Chemistry (3011)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- The outer electronic configuration ns^2np^4 corresponds to which one of the following elements in its ground state?
a. Fe b. Ca c. Cr d. S
- If an atom has 6 unpaired electrons, what is likely to be the total spin of the electron.
a. 3 b. 2.5 c. 2 d. 4
- The maximum number of electrons that can be accommodated in a sub shell for $l=3$ is
a. 2 b. 10 c. 6 d. 14
- If M-shell has 4 electrons, what would be the name of element?
a. Silicon b. Phosphorus c. Sulphur d. Beryllium
- Nascent hydrogen consists of
a. hydrogen atoms with excess energy
b. hydrogen molecules with excess energy
c. hydrogen ions in the excited state
d. solvated hydrogen atom
- Which of the following is peroxide?
a. NO_2 b. SO_2 c. C_3O_2 d. BaO_2
- Which is detected by Nessler's reagent?
a. NH_4^+ b. MnO_4^- c. PO_4^{3-} d. CrO_4^{2-}
- Which of the following does not acts as bleaching agent?
a. Cl_2 b. SO_2 c. I_2 d. H_2O_2

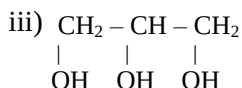
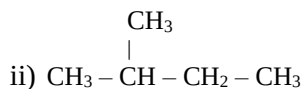
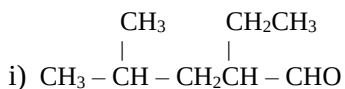
- $$\begin{array}{ccc} \text{CH}_3 & & \text{CH}_3 \\ & \diagdown \quad \diagup & \\ & \text{C} = \text{C} & \\ & \diagup \quad \diagdown & \\ \text{CH}_3 & & \text{CH}_3 \end{array}$$

- ### Group - B

[8 × 5 = 40]

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4. Halogens are extremely reactive non-metal and forms haloacid with hydrogen.
 - a. Write down the balanced chemical reaction for the preparation of HCl. [1]
 - b. Why can't HBr be prepared in the same way as HCl? [1]
 - c. Write alternative method for preparation of HBr. [1]
 - d. How does HI react with CuSO_4 solution? [1]
 - e. Mention one use each of Cl_2 and Br_2 . [1]
5. Hydrogen and oxygen are the abundant elements in the earth.
 - a. Write one chemical reaction to show that nascent hydrogen is better reducing agent than molecular hydrogen. [1]
 - b. Classify the following oxides with a suitable reason. [2]
 Cr_2O_3 , I_2O_5 ,
 - c. How CFCs deplete the ozone layer in stratosphere? [2]
6. Ammonia is the most valuable compound of nitrogen.
 - a. Define active nitrogen. [1]
 - b. What is ring test? Write reaction involved. [2]
 - b. What happens when ammonia react with oxygen in presence of catalyst? [1]
 - c. How ammonia react with ferric chloride solution? [1]
7. Periodic table is used to study different periodic properties.
 - a. Electron affinity of noble gas is zero. Give reason. [1]
 - b. Which of the following is the larger in size?
 Ne , Mg^{++} . Give an appropriate reason. [2]
 - c. Among the element B, Al, C, Si [2]
 - i) Which elements has highest first ionization energy.
 - ii) Which element has most metallic character? justify your answer in each case.
8. International Union of Pure and Applied Chemistry (IUPAC) developed a system of giving the name to an organic compound called IUPAC system.
 - a. Write the structure of the following IUPAC names, [2]
 - i) 3-methylbutan-2-one ii) 3-ethyl-2-methylhexane
 - b. Write the IUPAC names of the following compounds, [3]



Group - C

Give long answer to the following questions.

[3 × 8 = 24]

9. The "atom" was regarded as a small indivisible particle-the unit of matter in all chemical changes.
- Write the postulates of Bohr's atomic model. [4]
 - State the "Hund's Rule of Maximum Multiplicity". Write all the four-quantum numbers of valence electrons of copper. [1+1]
 - Name the quantum number which explain the orientation of orbitals and quantum number which explain main energy. Also, explain the quantum number which give information about shape of orbitals [2]
10. Nitric acid, also known as aqua fortis, is a highly corrosive mineral acid.
- How is ammonia manufactured by Haber's process? Write the favourable conditions for higher yield of ammonia by this process. [1+2]
 - Draw a flowsheet diagram of manufacture of nitric acid by Ostwald's process. [2]
 - What is passivity of iron? [1]
 - List out the oxyacids of nitrogen with their molecular formula and oxidation number of nitrogen. [2]
11. Organic chemistry is the branch of chemistry which deals with study of hydrocarbons and their derivatives.
- What do you mean by open chain and closed chain compounds? Give some examples to illustrate. [2]
 - What is Homologous series? Mention the characteristics of homologous series. [3]
 - Write the homologous series containing -CHO functional group upto four members with IUPAC name. [2]
 - What are heterocyclic compounds? Give one example of it. [1]

❧ - The End - ❧